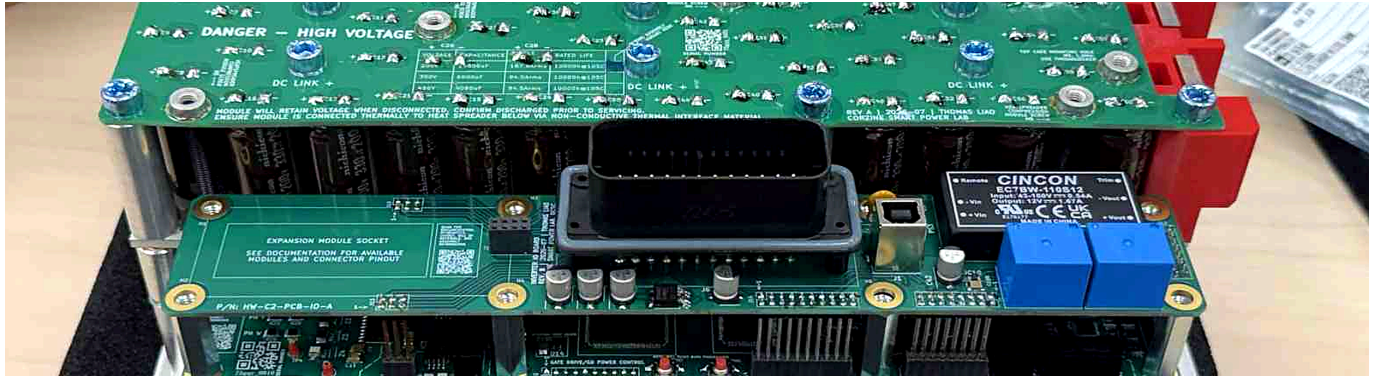


The open, safety-engineered power-conversion platform

Built on our bench-tested dual-MCU control architecture. Swappable power stages scale from 10 kW to the megawatt class for motor drives and DC/DC conversion, with SiC-ready designs.



Size 2 is a 200–450 V-class design rated 465 A continuous and 600 A peak for 60 seconds, with self-commissioning for PMSM and induction motors.

■ Built and bench-tested

Size 2 has completed a **180 V bring-up**, a **one-hour endurance run**, a **20-minute reversal stress test**, and instrument-verified motor self-commissioning. The Size 2 chassis is being tested for use on a **2018 Zero DSR electric motorcycle**.

■ Safety engineered

Six independent paths to the safe state combine hardware shutdown with main-controller and safety-coprocessor authority. The HARA and FMEA map to **ISO 26262** and **IEC 61800-5-2**, with hazards traced to safety requirements and validation.

■ Open everything

Schematics, firmware, safety analyses, assembly data, interactive telemetry, and **RTE Studio**, an open-source model-based development and code-generation tool, are public at docs.openvvf.org. Sponsors and researchers can inspect what was built and how it performed.

MODULAR POWER ARCHITECTURE

Reusable control, scalable power stages

CONTROL MODULE

Dual-MCU control · FOC · independent safety

POWER STAGE

Gate-driver boards · capacitor boards · IGBT or SiC modules



Size 1

Approximately 10 kW class for light EVs and scalable designs

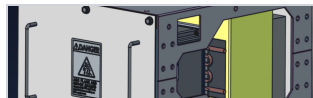
■ EARLY DESIGN



Size 2 inverter

200–450 V class · 465 A continuous design rating

■ IMPLEMENTED, UNDER TEST



Size 2 DC/DC

15 kW · 450 V class · reuses Size 2 boards and power stage

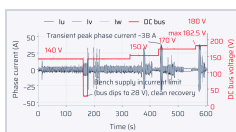
■ RELEASED DESIGN



Size 3

Up to 1,200 V · 1,400 A · modular liquid-cooled enclosure

■ IN DEVELOPMENT



Live telemetry

Ten published reports include raw, interactive logs for the browser-based Telemetry Viewer. Approximately 38 A peak current and clean recovery from a supply current-limit event.



SUPPORT THE SMART POWER LAB

Smart Power Lab contacts

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CURRENT SUPPORTERS

